

Lab2 Solution&Hint

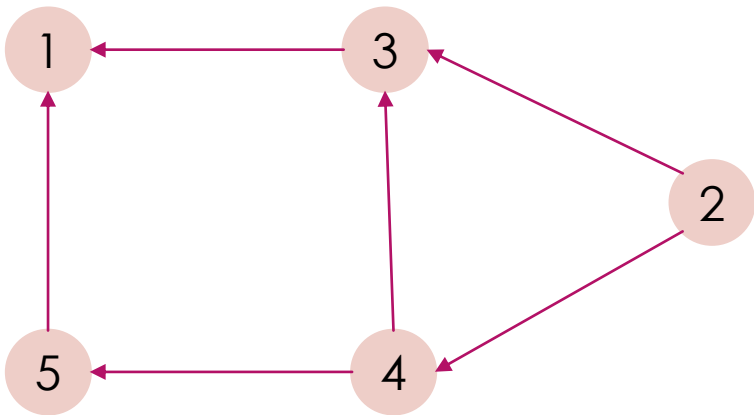
YAO ZHAO

Lab2.A: Food chain

- ▶ There are M predation relationships among N species in some fauna. It is guaranteed that there is no cyclic predation in the food web.
- ▶ A food chain is defined by a sequence of species $[a_1, a_2, \dots, a_k]$, where
 - ▶ nothing can prey on a_1
 - ▶ a_k can prey on nothing
 - ▶ a_i can prey on a_{i+1} for $i = 1, 2, \dots, k - 1$
- ▶ Two food chains are different, if and only if their sequences are different.
- ▶ Now each of these N species wants to know how many food chains involve with it, module $10^9 + 7$.

Sample Input 1

5 6
4 5
4 3
2 4
3 1
2 3
5 1



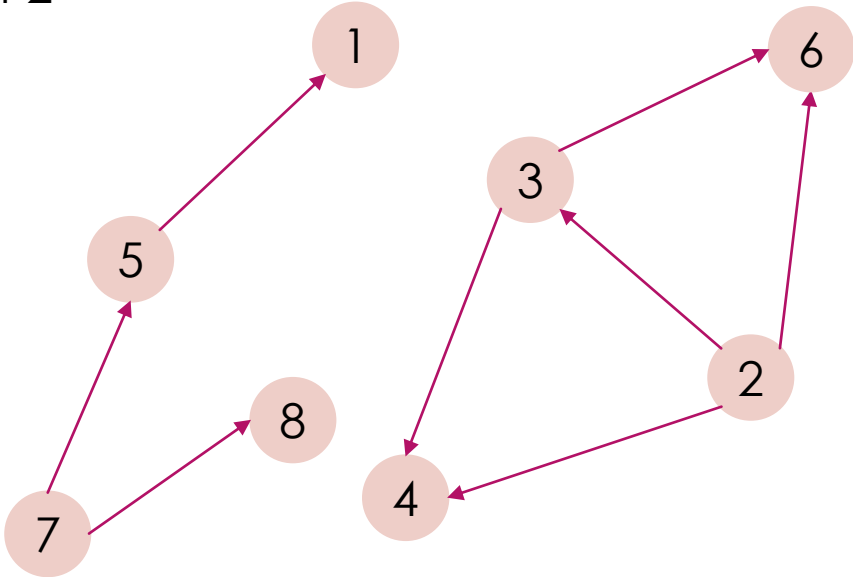
species	food chains	number
1	[2,3,1] [2,4,3,1] [2,4,5,1]	3
2	[2,3,1] [2,4,3,1] [2,4,5,1]	3
3	[2,3,1] [2,4,3,1]	2
4	[2,4,3,1] [2,4,5,1]	2
5	[2,4,5,1]	1



Sample Output 1
3 3 2 2 1

Sample Input 2

8 8
2 4
7 5
3 4
3 6
2 6
7 8
2 3
5 1



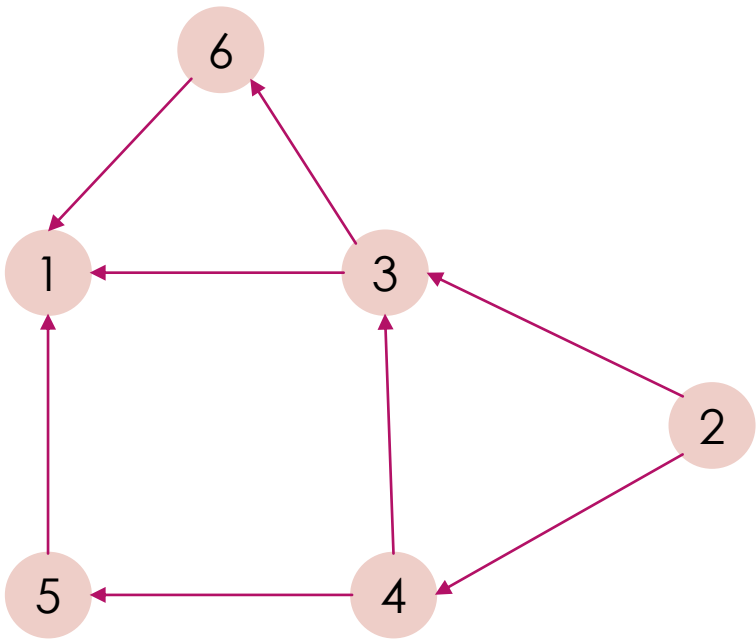
species	food chains	number
1	[7,5,1]	1
2	[2,4] [2,3,4] [2,6] [2,3,6]	4
3	[2,3,4] [2,3,6]	2
4	[2,4] [2,3,4]	2
5	[7,5,1]	1
6	[2,6] [2,3,6]	2
7	[7,5,1] [7,8]	2
8	[7,8]	1



Sample Output 2
1 4 2 2 1 2 2 1

Sample Input 3

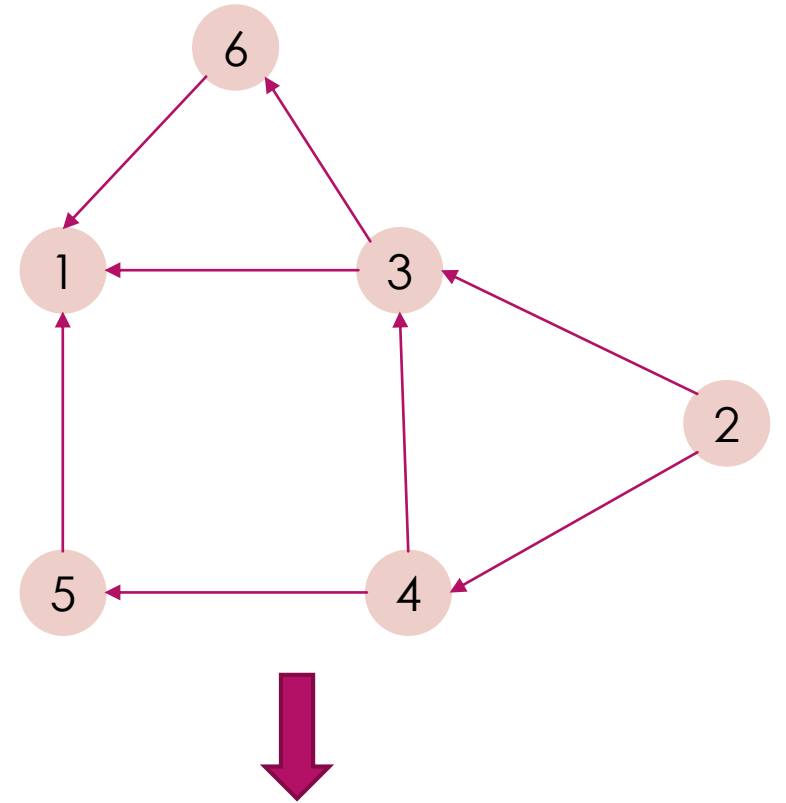
6 8
4 5
4 3
2 4
3 1
2 3
5 1
3 6
6 1



species	food chains	number
1	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5
2	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5
3	[2,3,1] [2,4,3,1] [2,3,6,1] [2,4,3,6,1]	4
4	[2,4,3,1] [2,4,5,1] [2,4,3,6,1]	3
5	[2,4,5,1]	1
6	[2,3,6,1] [2,4,3,6,1]	2

Try to find all topo order and update the chains number of each node

Find all nodes with no incoming edges
Init their chains number to 1 and put them to queue Q
While Q is not empty:
 get a node e from Q
 for each child c of node:
 $c.\text{indegree} \leftarrow c.\text{indegree} - 1$
 $c.\text{chainsnumber} \leftarrow c.\text{chainsnumber} + e.\text{chainsnumber}$
 if $c.\text{indegree} == 0$
 put c to Q



Why?



Can't find all chains number



1	2	3	4	5	6
5	1	2	1	1	2

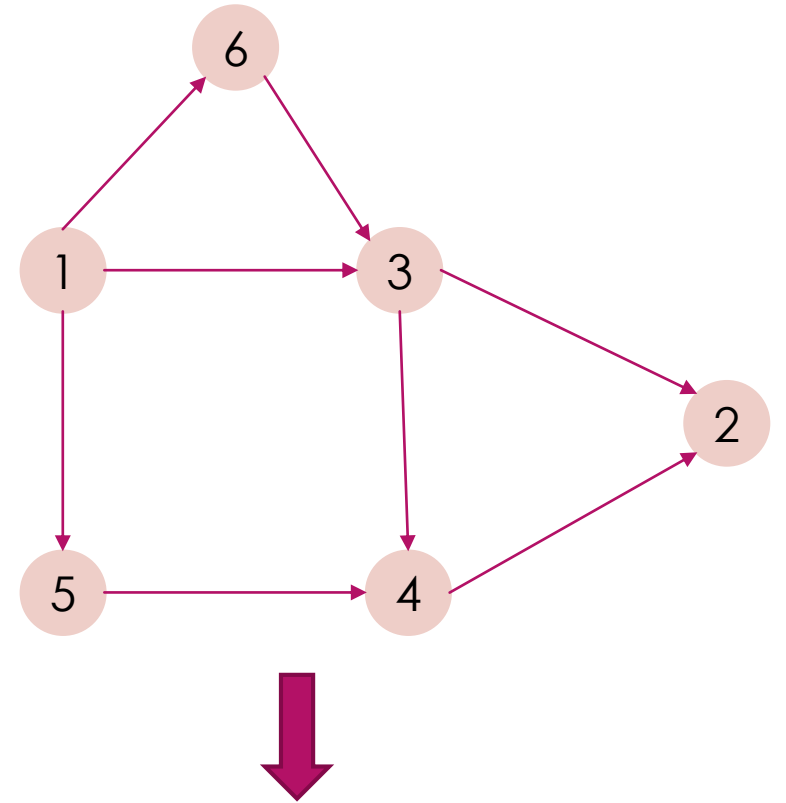
species	food chains	number	result
1	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5	5
2	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5	1
3	[2,3,1] [2,4,3,1] [2,3,6,1] [2,4,3,6,1]	4	2
4	[2,4,3,1] [2,4,5,1] [2,4,3,6,1]	3	1
5	[2,4,5,1]	1	1
6	[2,3,6,1] [2,4,3,6,1]	2	2

Only find the combination before the node

How to find all?

Reverse the edges and execute again:

Find all nodes with no incoming edges
Init their chains number to 1 and put them to queue Q
While Q is not empty:
 get a node e from Q
 for each child c of node:
 $c.\text{indegree} \leftarrow c.\text{indegree} - 1$
 $c.\text{chainsnumber} \leftarrow c.\text{chainsnumber} + e.\text{chainsnumber}$
 if $c.\text{indegree} == 0$
 put c to Q

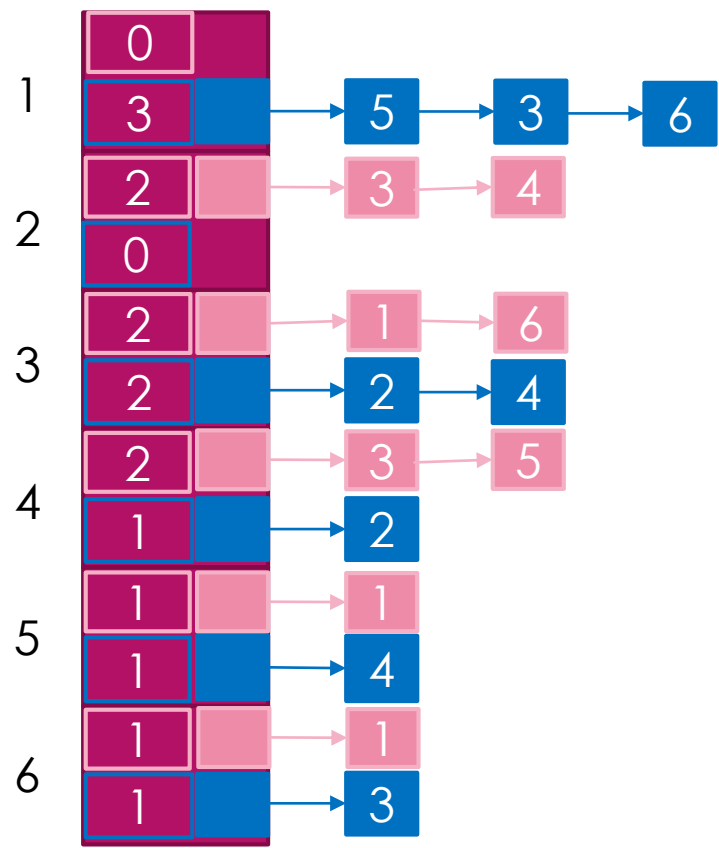
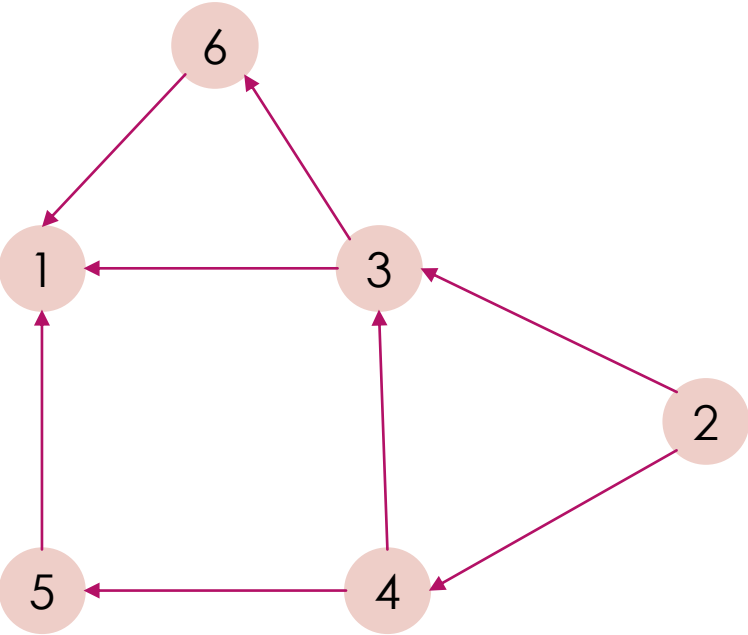


1	2	3	4	5	6
1	5	2	3	1	1

species	food chains	number	Result	Reverse-result
1	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5	5	1
2	[2,3,1] [2,4,3,1] [2,4,5,1] [2,3,6,1] [2,4,3,6,1]	5	1	5
3	[2,3,1] [2,4,3,1] [2,3,6,1] [2,4,3,6,1]	4	2	2
4	[2,4,3,1] [2,4,5,1] [2,4,3,6,1]	3	1	3
5	[2,4,5,1]	1	1	1
6	[2,3,6,1] [2,4,3,6,1]	2	2	1

number = result * (Reverse-result)

Recommended data structure



- Out degree
- Point to out edges list
- In degree
- Point to in edges list

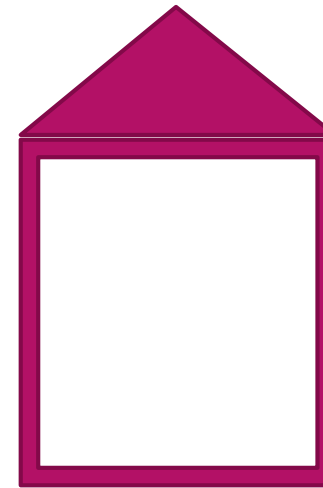
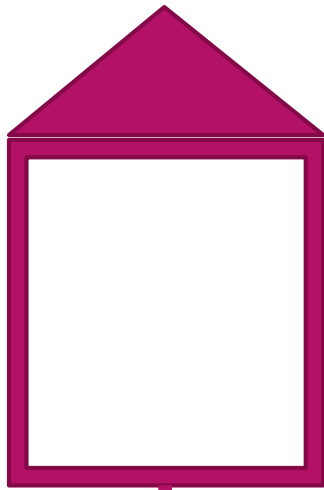
Lab2.B: Legendary Grabbing Machine

- ▶ Satori is a hunter who likes hunting bunnies.
- ▶ Initially at moment 0, there are N bunnies, the i^{th} of which is at position p_i .
- ▶ There are also M nests, the i^{th} of which is at position q_i . Each nest can hold at most C bunnies.
- ▶ Each bunny can move at most 1 unit of distance within 1 unit of time. Once a bunny enters some nest, it will be completely safe from Satori.
- ▶ Satori's **Legendary Grabbing Machine** takes T units of time to charge. Once the machine finishes charging at moment T , all bunnies that are out of nest will be captured. Note that bunnies entering nest at moment T will be safe.
- ▶ The bunnies very are united. They wish to know the maximal number of safe bunnies if they move optimally.



Input:
2 2 1 5
45 55
40 60

Legendary Grabbing Machine



40

45

50

55

60

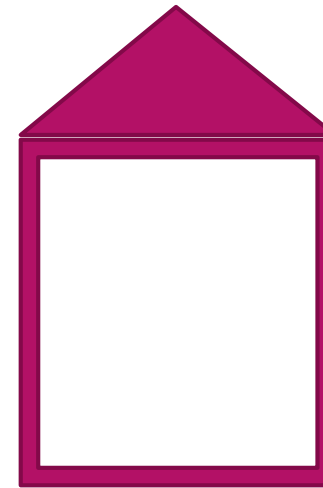
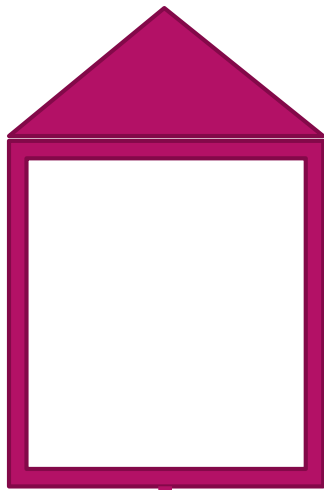
65



Input:
2 2 1 5
45 55
40 60

Output:
2

Legendary Grabbing Machine



40

45

50

55

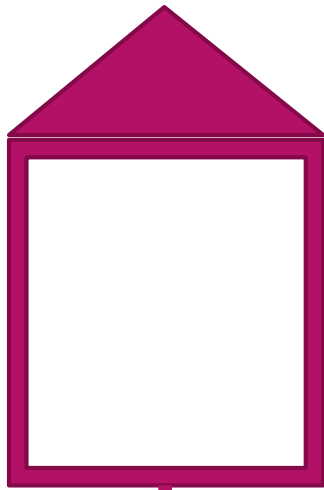
60

65



Input:
2 2 1 5
55 65
40 60

Legendary Grabbing Machine



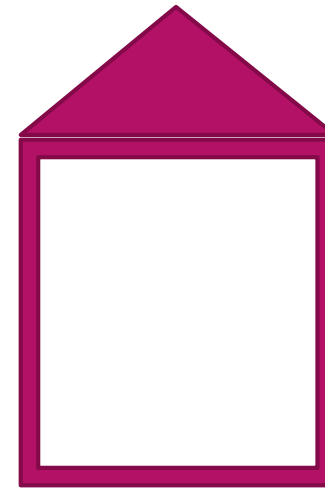
40

45

50



55



60



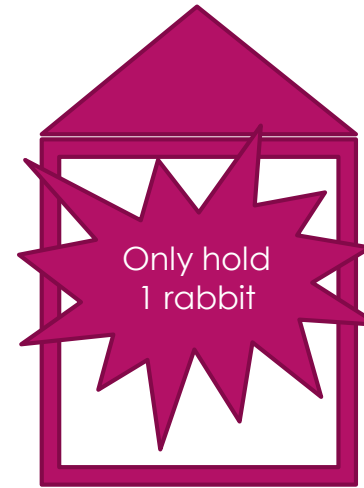
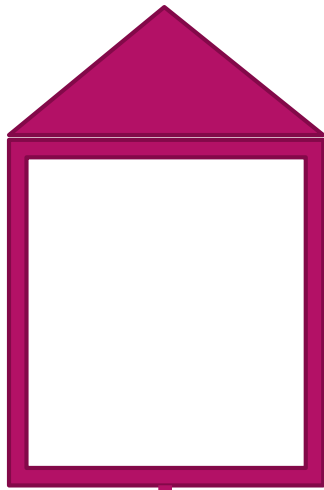
65



Input:
2 2 1 5
55 65
40 60

Output:
1

Legendary Grabbing Machine



40

45

50

55

60

65

Hint:

Sort the nests by position

Sort the bunnies by position

Start from one direction or another, greedily fill the nests